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Company of the Leister Group

MICRO-OPTICS | INFRARED SOURCES | MASS FLOW DEVICES | LASER GAS DETECTION

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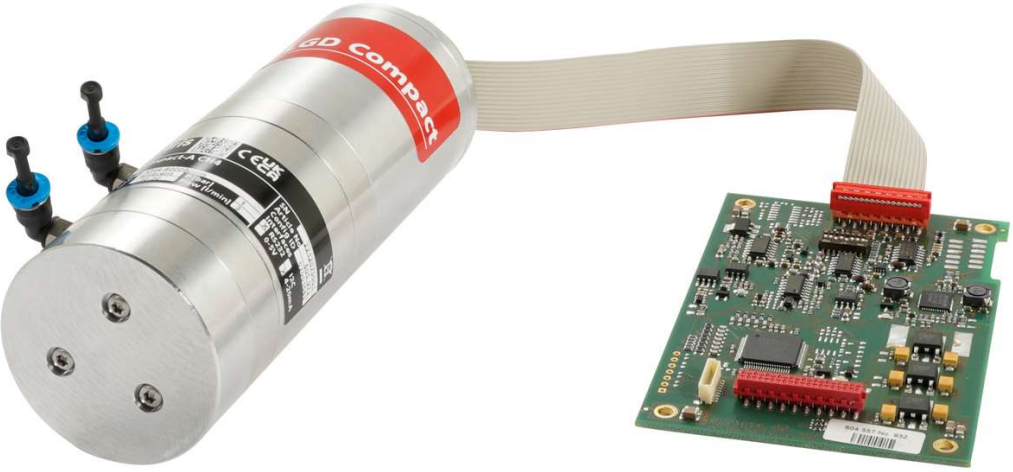

Operating & Integration Instructions

Laser Gas Detection
OEM Module
LGD Compact

Foreword

Dear customer, we thank you for acquiring an Axetris LGD Compact laser gas detection module. Please read this document carefully and entirely, to professionally install and operate your device.

Axetris laser gas detection modules are manufactured with extreme attention and the highest quality standards to offer you the best performance for full benefit in your application. Should you require any additional information or require troubleshooting, please contact us for support.

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Important safety instructions

Warnung		Vorsicht	
	Gefahr! Das Gerät nicht öffnen. Laser Klasse 1.		Das Gerät ist nur mit einem Druck von 800 – 1100 mbar zu betreiben.
	Das System darf nicht in einer explosiven Umgebung benutzt werden. Keine ATEX Zertifizierung.	Hinweis	
	Verbrennungsgefahr! Gasanschlüsse und Gehäuse nicht in heissem Zustand berühren. Gerät abkühlen lassen.		Reparaturen sind ausschliesslich durch eine autorisierte Axetris-Service-Stelle ausführen zu lassen.
	Beim Messen von potenzial gefährlichen Konzentrationen von giftigen Gasen muss der Auslass des Sensors in einen sicheren Bereich geführt werden.		Sämtliche in der Bedienungsanleitung gegebenen Informationen und enthaltenen Warnungen sind einzuhalten.

Warning		Caution	
	Danger! Do not open the device. Class 1 laser.		The device may be used within a pressure range from 800 to 1100 mbar only.
	The system may not be used in a potentially explosive environment. No ATEX certification.	Note	
	Risk of burning! Do not touch gas connections or housing when they are hot. Allow the device to cool down.		Repairs shall be assigned exclusively to an authorized Axetris service point.
	When potentially hazardous concentrations of toxic gases are measured, the sensor outlet must be transferred to a safe area.		All information and warnings set out in the Operating Manual must be observed.

Avertissement		Prudence	
	Danger ! Ne pas ouvrir l'appareil. Laser classe 1.		L'appareil doit uniquement être utilisé à une pression comprise entre 800 et 1 100 mbar.
	Le système ne doit pas être utilisé dans une atmosphère explosive. Pas de certification ATEX.	Remarque	
	Risque de brûlure ! Ne pas toucher les raccords de gaz ni le boîtier lorsqu'ils sont chauds. Laisser refroidir l'appareil.		Les réparations doivent exclusivement être effectuées par un centre de service Axetris agréé.
	Lors de la mesure de concentrations potentiellement dangereuses de gaz toxiques, la sortie du capteur doit être guidée vers une zone sûre.		L'ensemble des informations et avertissements fournis dans la notice d'utilisation doivent être pris en compte.

Advertencia		Precaución	
	Peligro No abrir el equipo. Láser clase 1.		El equipo solo debe operarse con una presión de 800 – 1100 mbar.
	No se debe utilizar el sistema en un ambiente potencialmente explosivo. Sin certificación ATEX.	**Indicación**	
	Peligro de quemaduras. No toque las conexiones de gas ni la carcasa si están calientes. Dejar que el equipo se enfríe.		Solo los puntos de servicio técnico Axetris están autorizados a realizar reparaciones en el dispositivo.
	Durante la medición de concentraciones potencialmente peligrosas de gases tóxicos debe colocarse la salida del sensor en una zona segura.		Deben respetarse todas las advertencias y la información incluidas en el manual de instrucciones.

Avvertenza		Attenzione	
	Pericolo! Non aprire l'apparecchio. Laser categoria 1.		Far funzionare l'apparecchio soltanto con una pressione di 800 – 1100 mbar.
	Il sistema non può essere utilizzato in un ambiente esplosivo. Nessuna certificazione ATEX.	**Nota**	
	Pericolo di scottature! Non toccare gli attacchi del gas e l'alloggiamento quando sono roventi. Fare raffreddare l'apparecchio.		Le riparazioni devono essere eseguite esclusivamente da un centro di assistenza autorizzato Axetris.
	Nella misurazione delle concentrazioni potenzialmente pericolose di gas tossici, lo scarico del sensore deve essere guidato in un ambiente sicuro.		Vanno osservate tutte le informazioni e tutti gli avvertimenti contenuti nelle istruzioni per l'uso.

警告		小心	
	危险！不可打开设备。激光级别 1。		只可以 800 – 1100 mbar 的压力来运行传感器。
	不允许在可引起爆炸的环境中使用此系统。无 ATEX 认证。	**提示**	
	灼伤危险！不要接触热的气体接口和外壳。冷却设备。		仅可由授权的 Axetris 服务部门进行维修。
	在测量有毒气体的潜在危险浓度时，必须将传感器的出口引至安全的区域。		必须遵守在操作说明书中给出的所有信息和警告。

警告		注意	
	危険！デバイスを開けないでください。クラス1のレーザー。		センサーは800～1100 mbarの圧力でのみ動作します。
	システムは、爆発性のある環境で使用することはできません。ATEX認定はありません。	注記	
	ヤケドの危険があります！高温状態において、ガス接続部およびハウジングに触れないでください。熱風融着機をクールダウンさせます。		認定されたAxetrisサービスセンターのみが修理を実施することができます。
	有毒ガスの潜在的に危険な濃度を測定する場合、安全な場所においてセンサーを差し込む必要があります。		取扱説明書で提供されているすべての情報および警告情報に従う必要があります。

경고		주의	
	위험! 장치를 개방하지 마십시오. 1 등급의 레이저.		센서는 반드시 800 – 1100 mbar의 압력으로 가동해야 합니다.
	본 시스템은 폭발 환경에서 사용하지 말아야 합니다. ATEX 인증 안됨.	주의사항	
	화상위험! 가스 연결부 및 하우징은 가열된 상태에서 접촉하지 마십시오. 이런 경우 장치를 냉각시키십시오.		수리는 반드시 인증된 Axetris 서비스 대리점을 통해 실시해야 합니다.
	잠재적 위험이 있는 유독 가스 농도를 측정하는 경우, 센서의 출구가 안전한 영역에 위치해야 합니다.		사용설명서에 명시된 모든 정보 및 여기에 포함된 경고 사항의 내용을 준수해야 합니다.

About this Operating & Integration Instructions

This Operating & Integration Instructions is the original and legally correct version.

The aim of this Operating & Integration Instructions is to provide the user all the necessary information to install, operate and maintain LGD Compact Series Products.

This Operating & Integration Instructions must be read by all individuals that will be using or servicing this device. System maintenance and service needs to be performed by qualified and Axetris AG approved personnel only. The appropriate handling of the product is required for trouble-free operation. This product is a high-precision measurement device. Careful installation and handling is necessary to achieve and maintain the specified measurement precision. Correct set-up and operation of the instrument will allow you to use it safely and to obtain reliable measurement results.

Any warranties regarding this product, and as mentioned in the *General Terms and Conditions of Axetris AG,* are void in the event the system is altered or damaged by the user, not installed, used and serviced according to this document.

Important notes in the text:

Throughout the text, warning and information notes in red, as shown below, inform the user about crucial points to observe. These warnings and informational notes must be followed scrupulously and Axetris AG takes no responsibility for any personal injury, death or any damage to equipment or property resulting from not observing these indications.

⚠ The warning sign indicates that severe personal injury/death and/or property damage might result from not observing safety instructions.

ⓘ The information sign points to particularly important information and instructions.

CAUTION

Product damage

- Read all instructions carefully before using the device.
- The LGD Compact gas sensor OEM-modules are not designed to sense liquid flow and damage will result if liquid is passed through the sensor.
- Gas loaded with particles will eventually contaminate the optics and affect the performance of the sensor. Please consult our Customer Support accordingly.
- The appliance must not be used in damp or moist surroundings.
- Only use accessories indicated in the instructions for use or which are recommended by the manufacturer.
- Failure to comply with these instructions can result in product damage.

Danger to life

- Unprofessional gas handling can cause injury or death. The use of the gas sensor modules should only be performed by qualified personnel.
- Do not use this product as a safety or emergency stop device or in any other application where failure of the product could result in personal injury or death.

IF THE TEMPERATURE OF THE GAS PROVIDED IS HIGHER THAN THE CELL, A HIGH RISK OF CONDENSATION AND IRREVERSIBLE DAMAGE OF THE CELL EXISTS.

DO NOT OPEN THE LGD COMPACT MEASUREMENT HEAD: LASER CLASS 1 INSIDE
(Peak Power 2.5 mW - Wavelength: 1500 - 2000 nm).

CLASS 1 LASER PRODUCT
(classified according to SN EN 60825:1:2014-08)

DO NOT USE THE SYSTEM IN AN EXPLOSIVE ATMOSPHERE: NOT ATEX CERTIFIED.

DO NOT USE WITH EXPLOSIVE GAS MATRIX: NOT ATEX CERTIFIED.

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1 GENERAL USER INFORMATION

1.1 LGD Compact introduction

Axetris' Laser Gas Detection OEM-modules (LGD), "LGD Compact", is a system for the continuous monitoring of gases such as CH_4, C_2H_6, C_2H_2, NH_3 and CO_2. Based on state-of-the-art *Tunable Diode Laser Spectrometry* (TDLS), the system has a very high selectivity to the target gas and features an innovative, patented measurement principle without the need for a physical reference channel. The system is based on a flow-through cell for extractive gas measurements at ambient temperature.

Typically, one system monitors one gas. Concentration ranges are customer-specific and can cover 0-100 ppm just as well as percentage range, depending on the gas and measurement conditions.

Calibration is factory set according to customer application requirements - generally re-calibration is not necessary except in cases where the customer application conditions differ significantly from the feasible factory calibration conditions. Offset and span adjustments can be performed on a maintenance basis and might be necessary due to systematic differences in calibration and gas delivery equipment. This can be done via the LGD Customer Frontend available on the Axetris` homepage.

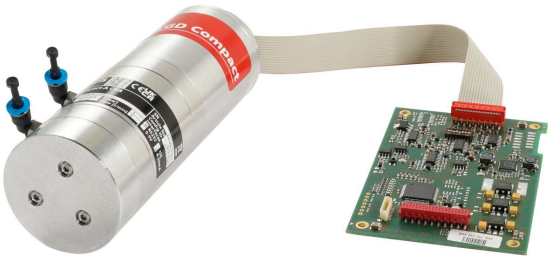


Figure 1 LGD Compact Series Generation 2

1.2 General Specifications

The LGD Compact Series Generation 1, in the phase out process until end of 2023 is progressively replaced by the Generation 2, that brings some integration improvements.

This document describes both Generations focusing on the gaps between both versions.

1.2.1 Environmental conditions

Parameter	Unit	Value / Range
Usage	-	Interior use
Operating temperature range	°C	-10 …+65[1]
Operating humidity	% r.H.	0 … 99, non-condensing
Operating pressure	mbar	800 … 1100
Storage temperature	°C	-40 … +80
Storage humidity	% r.H.	0 … 99, non-condensing

[1] For application specific temperature ranges please see the specifications in the LGD Compact Datasheet

1.2.2 Mechanical characteristics

Parameter	Unit	Value / Range
Measurement cell	-	Flow-through set-up
Cell volume	ml	19
Gas flow (min - max)	ml / min	100 - 3'000

Dimensions of sensor module	mm	Generation 1
		163 (length)
		50 (diameter w/o fittings and electrical connector)
		Generation 2:
		Basic: 146 (length) // 50 (diameter)
		EMC Conform: 163 (length) // 50 (diameter)

Approx. weight sensor module	g	Generation 1 ≤ 600 Generation 2: Basic: 510 EMC Conform: 630
Dimensions of electronic unit with housing	mm	Generation 1 257 x 83 x 26 (L x W x H) Generation 2: 163 x 78 x 43 (L x W x H))
Dimensions of electronic unit without housing (optional)	mm	105 x 70 x 11 (L x W x H, main board) 84 x 70 x 11 (L x W x H, interface board)

1.2.3 Electrical characteristics

Parameter	Unit	Value / Range
Voltage supply with interface board; min. - max.	VDC	10 - 30 (max. ripple ± 100mV, max. current 1A, inrush current limitation)
Voltage supply without interface board; min. - max.	VDC	3.7 – 5 (no inrush current limitation)
Power consumption	W	≤ 1 at T_{amb} 20°C (Up to 3 W in extreme conditions)
Start-up time	s	≤ 30

1.2.4 Data interface

Module powered at 10 - 30 VDC (with interface board)

Parameter	Unit	Value / Range
RS 232 EIA	-	RS232 protocol
Analog Current Output Analog Voltage Output	mA V	4 - 20, 12-bit resolution 0 - 5, 12-bit resolution
Resistive loads for voltage output for current output	 kΩ kΩ	 > 2 < 0.47
Capacitive loads For voltage output For current output	 pF pF	 < 200 < 200
Digital Alarm Outputs (relay)	-	Relay power supply: 10 V < Vcc Relay < 30 V; max. 0.75 A Various alarm outputs available, for more details refer to chapter 2.3
Protection		ESD protected

Module powered at 3.7 – 5 VDC (without interface board)

Parameter	Unit	Value / Range
RS 232 TTL	-	RS232 protocol, TTL signal amplitude 0 - 3.3 VDC
Analog Voltage Output	V	0 - 2.5, 12-bit resolution
Resistive loads	kΩ	> 5
Capacitive loads	pF	< 100
Digital Alarm Outputs	-	TTL signal amplitude 0 – 3.3 VDC Note: contact Axetris if needed
Protection		No protection implemented

1.3 Measurement software / Customer Frontend

There is a measurement software / Customer Frontend available for download on the Axetris` homepage.

https://www.axetris.com/en

The LGD Customer Frontend is a .NET application providing a user interface to LGD modules using the LGD communication driver.

Users have access to normal LGD functions including reading measurements and diagnostic data and setting a defined subset of parameters. To provide technical support extended functions are granted to Technical Support Engineers having access to the restricted LGD Customer Frontend Secure Access application.

The application is deployed as a .NET Core self-contained single executable file. No special installation process is needed. The user only needs to download the file, store it locally and execute it. It is also possible to store the file on a central file server and execute it from there.

Further information about the Frontend software itself and the handling can be found under "Help" menu in the LGD Frontend.

2 SYSTEM SET-UP FOR EVALUATION

2.1 Operation conditions, storage & transport

The instrument must be used within the specified ambient conditions and temperature range. Depending on information provided by the customer, the factory calibration temperature range might cover a narrower range adapted to the application and reducing the complexity of the calibration. Please find conditions for operation and storage in chapter 1.2.1.

Dust or condensing water in the measurement cell leads to signal reduction of the device. It is therefore crucial to ensure that the measurement gas flow is free of dust and that condensation cannot occur.

⚠ The systems are not designed for use in an explosive atmosphere.

ⓘ Irreversible system damage might occur if the use of adequate and application-specific gas filtration technology is omitted. This is an optical measurement system. For further information on gas conditioning and filtering please refer to 2.5.

ⓘ Relative humidity in the measurement gas and measurement conditions must be such that no condensation occurs inside the measurement cell.

ⓘ Each sensing module is calibrated to a certain gas temperature and pressure. Deviations in both parameters can lead to significant degrading of measurement accuracy.

The LGD Compact module continuously measures the gas concentration in the measurement cell. Default measurement time is roughly 1 s (user changeable to up to 600 s integration time or “averaging”). The LGD Compact has a start-up time of 30 s to give a correct concentration reading.

For any shipments of the system, it is recommended to retain and re-use the original shipment material, which has been tested and proven to be effective protection. Relevant Operation & Storage conditions as described above as well as in the Product Datasheet need to be respected for shipment. In case of a return Axetris AG will not be responsible for shipping damage because of improperly packaged systems and recommends customers to ship systems insured for full value.

2.2 System dimensions and mounting

⚠ Do not open the system. Opening might lead to irreversible damage.

⚠ Prevent any dust or other material from entering the measurement cell.

This chapter describes the dimensions of the sensor module, of the electronics housing and the printed circuit boards inside the electronics housing. Further described are the mounting holes of the sensor module.

2.2.1 Generation 1

2.2.1.1 Sensing module

The LGD Compact-A module has 2 x M4 threaded mounting holes located on the side (see **Figure 2,** recommended to use two pan head cap screw M4x16 (1.5 Nm) for mounting). For gas supply there are straight/angled QSML-M5-4 quick connectors for 4 mm diameter tubes.

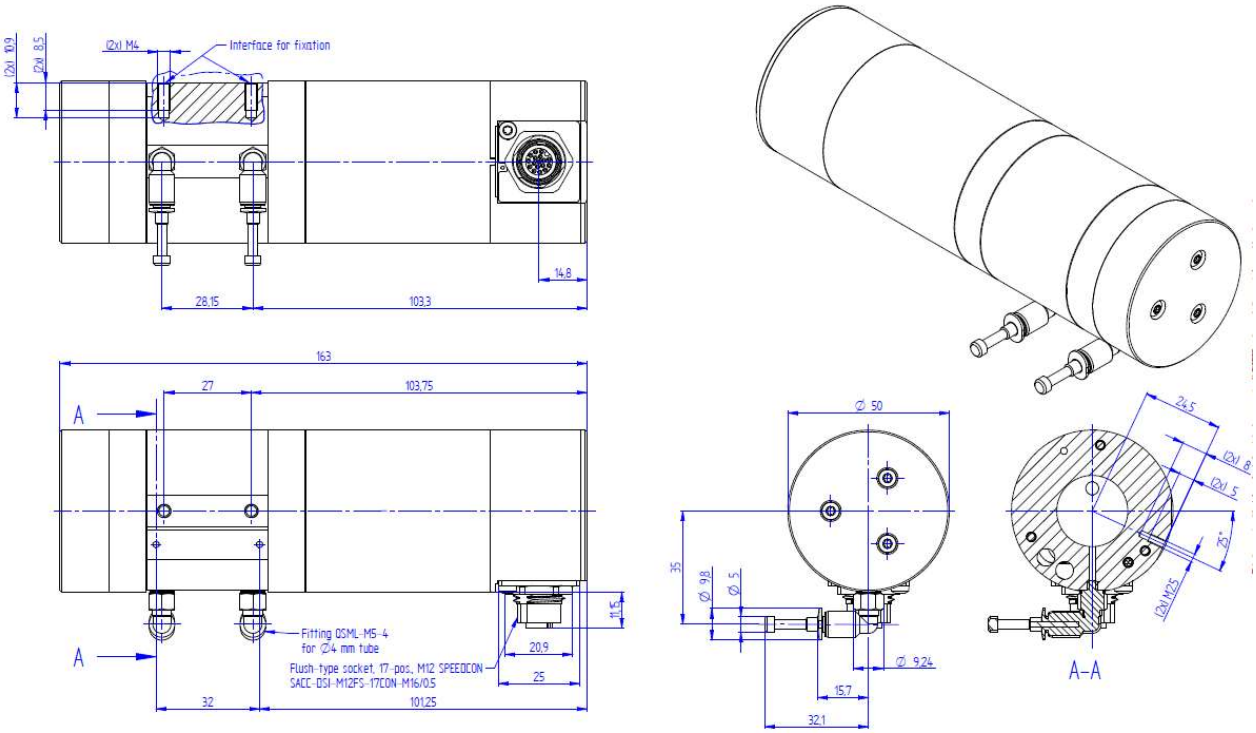


Figure 2 LGD Compact-A module dimensions and interfaces for fixation and gas supply

2.2.1.2 Electronics

The electronics is in a separate housing.
The outer dimensions of the electronic housing are 257 x 83 x 26mm (LxWxH) without connectors and screws.
To avoid damage during operation on the electronics it`s recommended to fix the electronics box with 4 pan head cap screws M4x10 (1.5 Nm) at the ends.
The electronics housing contains two printed circuit boards (PCBs) as shown in **Figure 3**.

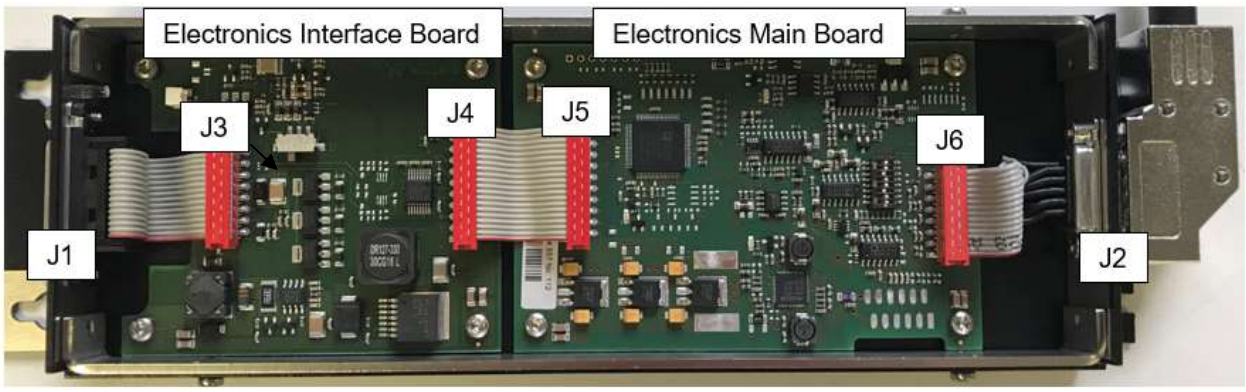


Figure 3 Electronics housing LGD Compact-A: interface board (left), main board (right)

The electronic main board should be connected to the sensor module just with the original cable delivered (recommendations for an own cable design can be seen in chapter 3.1.4).

The right position for connecting the cable to the sensor module are marked on the cable connector as well as on the sensor module. To fix the cable, the wheel must be screwed counter clockwise.

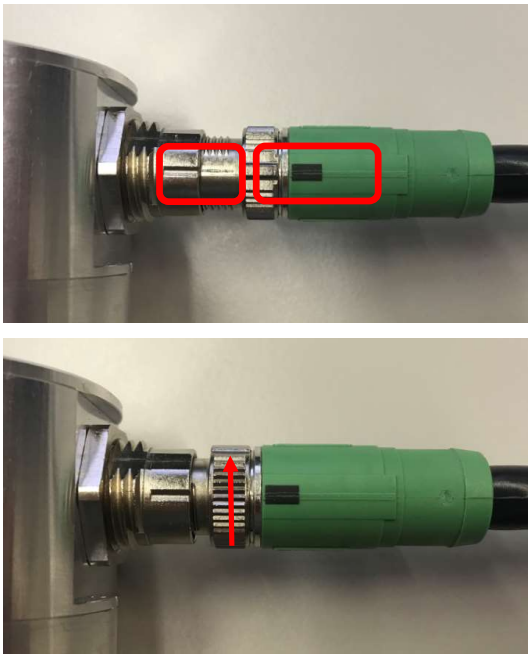


Figure 4 Correct connection of cable and sensor module

2.2.2 Generation 2

Same as for Generation 1, the LGD Compact-A module Generation 2 has 2 x M4 threaded mounting holes located on the side (use two pan head cap screw M4x16 (1.5 Nm) for mounting). For gas supply there are straight/angled QSML-M5-4 quick connectors for 4 mm diameter tubes

EMC conform product execution

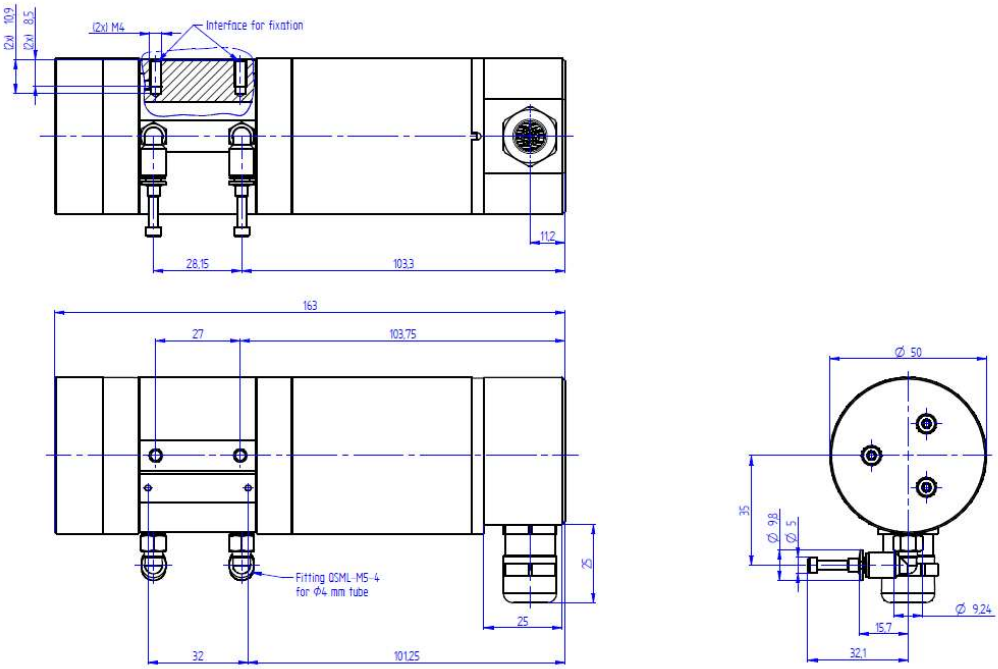


Figure 5 Generation 2 LGD Compact-A module EMC Conform execution

dimensions and interfaces for fixation and gas supply

Remark: Generation 2 has a stappled boards inside the housing, but connections are the same as Generation 1.

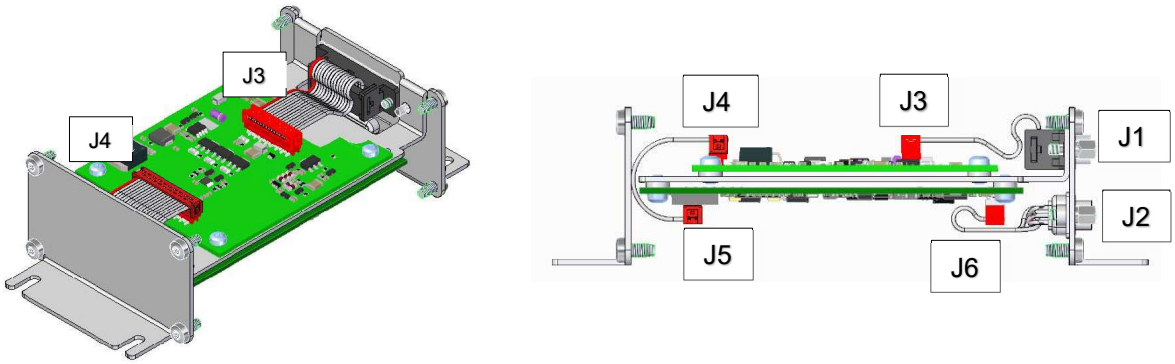


Figure 6 Electronics housing LGD Compact-A Generation 2: interface board (up), main board (down)

Basic product execution

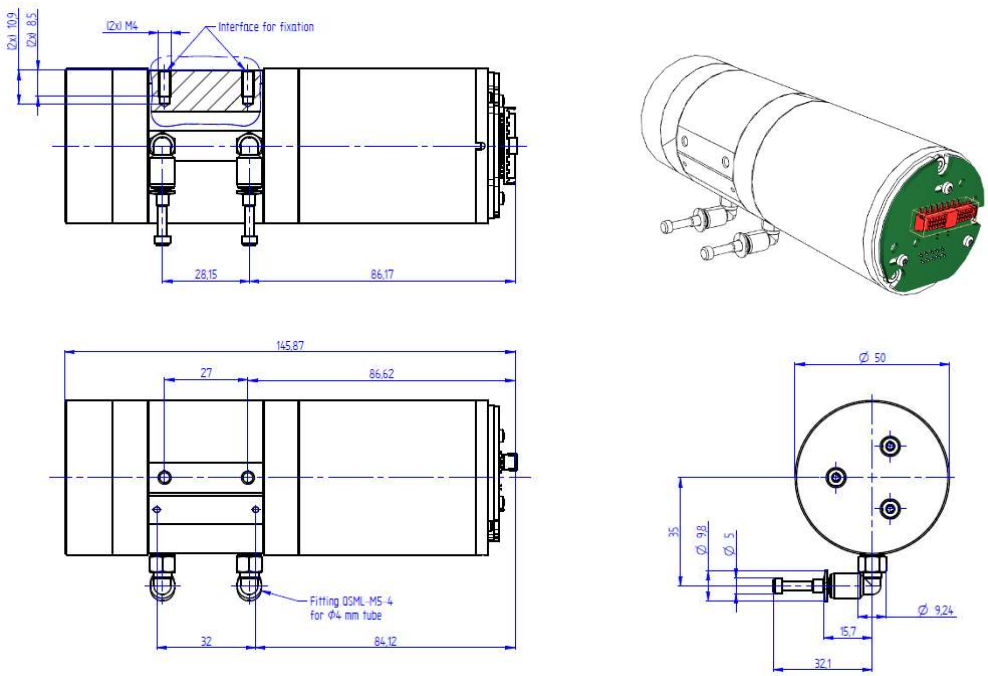


Figure 7 Generation 2 LGD Compact-A module Basic Execution dimensions and interfaces for fixation and gas supply

the sensor module is connected to the main electronic board via a Flat Ribbon cable.

The pin description of Micromatch connector (J7) can be found in Chapter 2.4.4

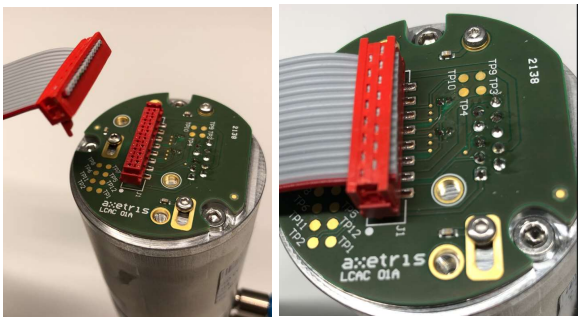


Figure 8 Generation 2 LGD Compact-A module Basic Execution / how ot connect flat band cable on sensing module

2.3 Electronic boards

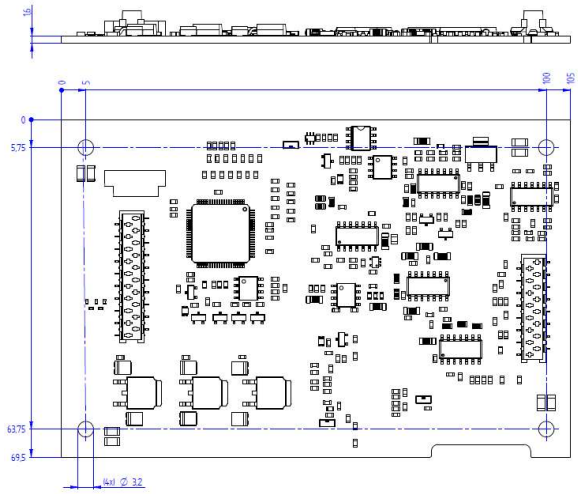


Figure 9 Dimensions of the main board

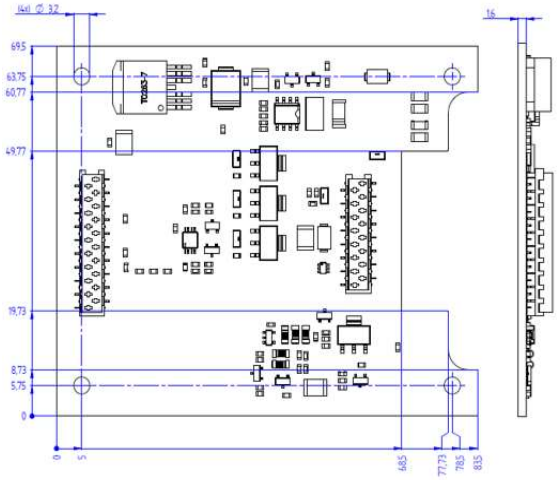


Figure 10 Dimensions of the interface board

2.4 Electrical Interfaces

The standard LGD interface is a RS232 connection. Connection to a USB port is possible via an RS232/USB adapter supplied with the system. Analog 4-20 mA or 0-5 V interfaces will be factory calibrated upon customer request. More details can be also found at chapter 1.2.4.

ⓘ To connect the system via a USB port please only use our tested RS232/USB adapter that is provided with the system.

ⓘ Please avoid starting-up or restarting a computer when the LGD is connected to the serial interface or through a USB port.

Position the LGD Compact module at the location where the measurement will be conducted and where computer, power supplies and gas delivery systems can be installed. For customer support and system service it is ideal if the computer also can be connected to the internet.

To allow simplified connectivity for evaluation a cable integrating both, system power supply and RS232, are provided (straight RS232 version, not crossed). The DB15 plug on the right-hand side in Figure 7 connects to

the mainboard of the instrument (mainboard connector in Figure 12), while the DB9 connector on the left-hand side in connects to the computer.

The electronic housing has two 15pol D-Sub connectors, one with two rows used for power supply and communication (J1) and the other one (three rows) for connection to the sensor module (J2).

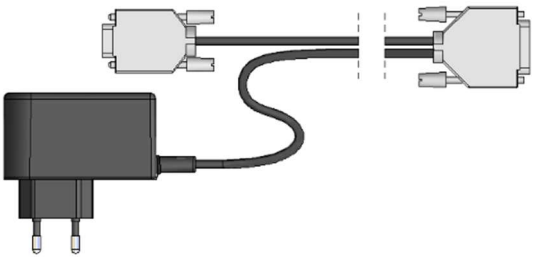


Figure 11 Combined power supply and RS232 cable. For integration purposes the connectors can be opened for OEM specific wiring

2.4.1 D-Sub power supply and communication J1 (15pol, 2 rows)

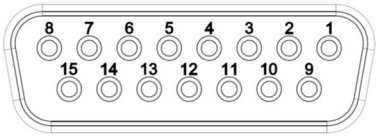


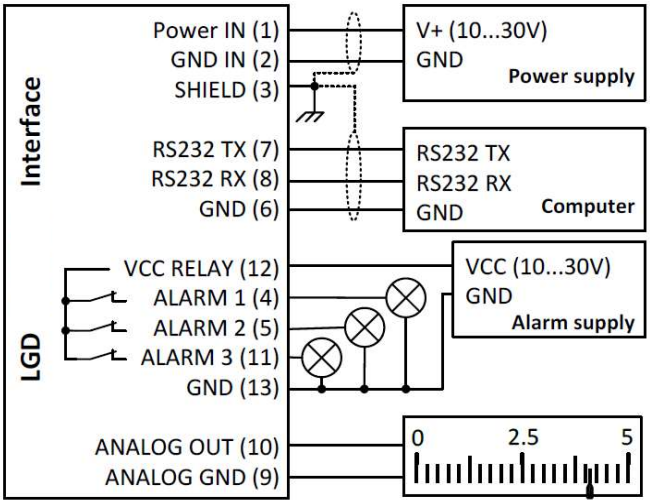
Figure 12 Connector assignment system power supply and user interface terminal

Figure 13 Connector assignment schematic

Ground Pin 6 & 13 do have the same potential.

UART Settings (default):

Baudrate:	9`600
Data bits:	8
Parity:	no
Stop bit:	1
Flow control:	no



	Pin	Name	Description
Power Supply	1	POWER IN	Power supply input 10 …30V
	2	GND IN*	Power supply ground (must be used separated from GND to ensure a proper power supply noise filtering)
	3	SHIELD	D-Sub 15 Connector – Housing
Digital Interfaces	6	GND*	Ground serial communication
	7	RS232 TX	UART serial communication
	8	RS232 RX	UART serial communication
Digital Alarms	4	ALARM 1	On/Off signal at defined conditions for relay control, hardware watchdog, etc.

	5	ALARM 2	On/Off signal at defined conditions for relay control, hardware watchdog, etc.
	12	VCC RELAY	Relay power supply: 10 V < VCC Relay < 30 V; max. 0.75 A
	13	GND*	Ground, digital alarms
Analog Interface	9	ANALOG GND*	Analog ground
	10	ANALOG OUT	Analog output Factory setting: output current (4-20mA); output voltage (0-5V) on request only
Not used	11, 14, 15	n/a	n/a
Shield	Case	SHIELD	Connected from D-Sub casing to cable shield

* Same potential

Table 1 Connector assignment 15pol D-Sub (2 rows)

2.4.2 Micromatch connector J3 (16pol)

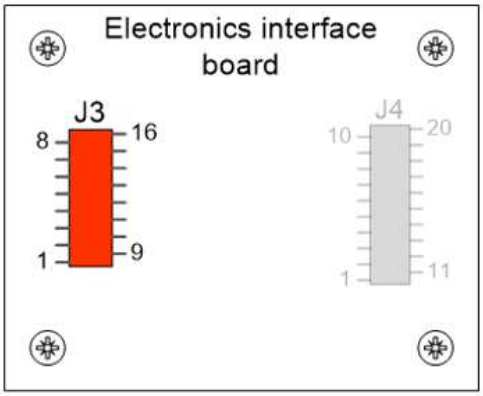


Figure 14 Connector assignment Micromatch 20pol

Ordering number receptacle:

- Manufacturer: TE connectivity
- Description: MICRO-MATCH MOW.16P
- Article number: 8-215083-6

The connector assignment of J3 is identical to J1 (see Table 1Table 1 Connector assignment 15pol D-Sub (2 rows)) except the additional pin 16 which is not used (see Table 2).

Table 2 Connector assignment 16pol Micromatch additional pin 16

	Pin	Name	Description
Not used	16	n/a	n/a

2.4.3 Micromatch connector J5 (20pol)

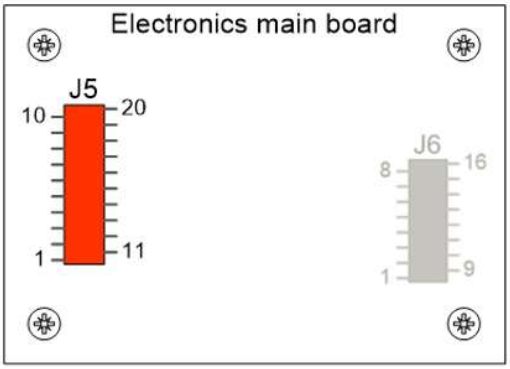


Figure 15 Connector assignment Micromatch 20pol

Ordering number receptacle:

- Manufacturer: TE connectivity
- Description: MICRO-MATCH MOW.20P
- Article number: 9-215083-0

⚠ Be careful in case of connecting J5 directly because there is no protection circuit implemented on the main board! The protection circuit is placed on the interface board.

Table 3 Connector assignment 20pol Micromatch

	Pin	Name	Description
Power Supply	1	5V*	Power supply input for mainboard
	2	5V*	Power supply input for mainboard
	3	3.3V	Power supply output 3.3V from mainboard to drive digital hardware on interface board; not used in case of removing the interface board
Digital Interfaces	4	TXD TTL	UART serial communication TTL-Level
	5	RXD TTL	UART serial communication TTL-Level
Digital Alarms	13	ALARM 1	Signal for alarm 1 from Microcontroller, contact Axetris if needed
	15	ALARM 2	Signal for alarm 2 from Microcontroller, contact Axetris if needed
Analog Interface	10	DAC ANALOGOUT	Analog output signal (0…2.5V) from Microcontroller, contact Axetris if needed
	16	HW INTERFACE PCB	Input for identification of interface board, connect to GND in case of removing the interface board
	11	GND**	Ground
	12	GND**	Ground
Not used	6…9	n/a	n/a
	14, 17…20	n/a	n/a

* Connected on main board
** same potential

2.4.4 Micromatch connector J7 (16pol)

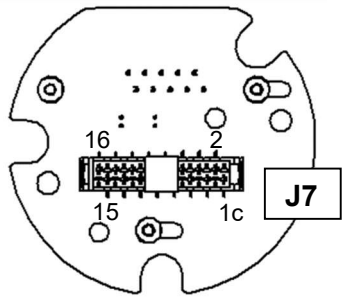


Figure 16 Connector assignment Micromatch 16pol

Ordering number receptacle:

- Manufacturer: TE connectivity
- Description: MICRO-MATCH MOW.16P
- Article number: 8-338069-6

Table 4 Connector assignment 16pol Micromatch

	Pin	Name	Description
Measurement signal	6	PD_I	Photodiode
	5	PD_GND*	
Laser control	1	Laser_Drive	Laser diode
	2	Laser_GND*	
	9	Laser_Supply3V3	
	15	Laser_U_{diff}	
	11	Laser_ID	
	3	Laser_Thermistor	Thermal resistance for laser temperature measurement

	8	Laser_GND*	
	4	Laser_TEC+	Peltier element for laser temperature stabilization
	13	Laser_TEC-	
	7	Laser_PT1K	Thermal resistance for optical head temperature measurement
	16	Laser_GND*	
Temperature measurement	10	Gascell_PT1K	Thermal resistance for gas temperature measurement
	12	Gascell_GND*	
Not used	14	Reserve	n/a

* Same potential

2.4.5 D-Sub sensor module J2 (15pol, 3 rows)

The sensor module is connected to the electronic housing via a three row D-sub connector. Figure 17 shows the pin assignment.

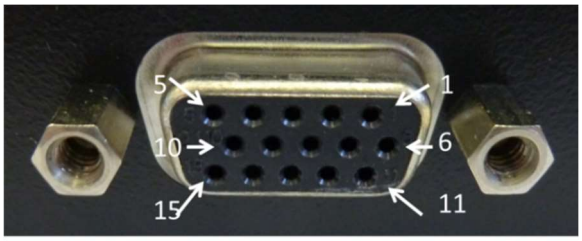


Figure 17 Sensor module connector assignment

Table 5 Connector assignment 15pol D-Sub (3 rows)

	Pin	Name	Description
Measurement Signal	11	PD_I	Photodiode
	12	PD_GND*	
Laser control	1	Laser_Drive	Laser diode
	9	Laser_GND	
	5	Laser_Supply3V3	
	8	Laser_U_{diff}	
	6	Laser_ID	
	2	Laser_Thermistor	Thermal resistance for laser temperature measurement
	3	Laser_GND*	
	10	Laser_TEC+	Peltier element for laser temperature stabilization
	7	Laser_TEC-	
	4	Laser_PT1K	Thermal resistance for optical head temperature measurement
Temperature measurement	13	Gascell_PT1K	Thermal resistance for gas temperature measurement
	14	Gascell_GND*	
Not used	15	Reserve	n/a
Shield	Case	SHIELD	Connected from D-Sub casing to cable shield

* Same potential

2.4.6 Supply voltages and output signals

Table 6 Supply voltages and output signals with/without interface board

	Supply Voltage	RS232 Communication		Analog Output	
		Low	High	Low	High
With interface board	10-30V	-12V	+12V	0V 4mA	5V 20mA
Without interface board	3.7 - 5V (Pmax = 1.6W)	0...0.4V (TTL)	2.4…3.3V (TTL)	0	2.5V

2.4.7 Load limitation for the analog interface

Since the system power supply can vary in output from 10V to 30V, it is important to consider of the load limitation of the 4-20 mA interface as displayed in Figure 18.

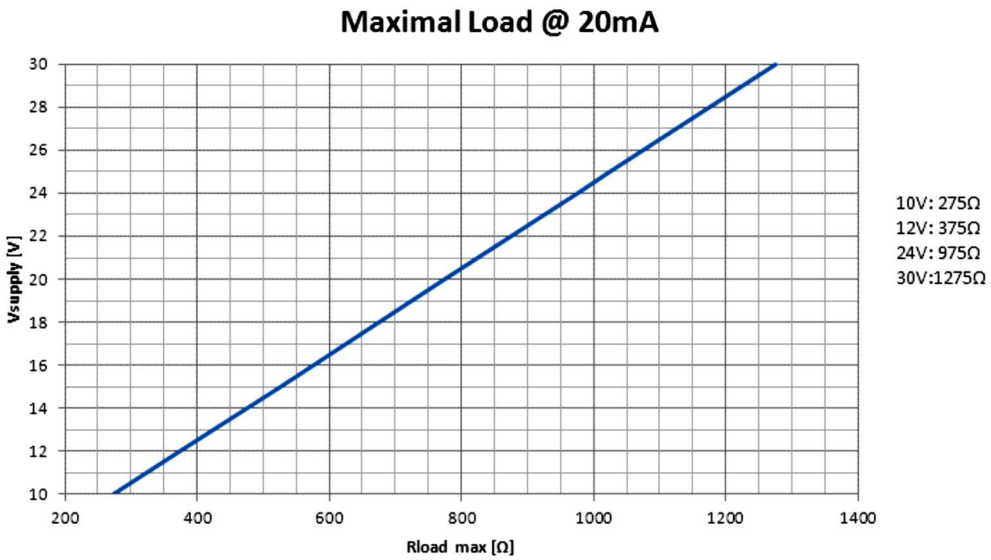


Figure 18 Load limitation of the 4-20 mA interface due to variable power supply (10V – 30V)

Graph: V_{supply} [V] versus R_{load} max [Ω]

2.5 Gas conditioning and filtering information

The LGD Compact Gas Sensor OEM-modules are based on an optical measurement technique where performance is linked to optical transmission through the measurement cell. Sample gas containing particles and aerosols are likely to contaminate the optical windows of the measurement cell and degrade the optical transmission more or less quickly. Contamination strongly depends on particle size, material, charging effects, etc. which can vary across different applications. It is therefore generally difficult to give a precise specification concerning the purity of the gas to avoid contamination on a reasonable time scale.

Due to the multitude of applications for the LGD, gas conditioning and filtering has to be carried out by the integrating Original Equipment Manufacturer (OEM). The following points are crucial to operate the module correctly and to achieve long operating lifetime:

- The sample gas must be dust-free; we recommend the use of appropriate filters with pore size ≤ 2µm.
- The sample gas must be moisture-free; we recommend the use of appropriate sample gas dryers ensuring r.H. ≤ 50% or non-condensing.
- The overall performance of the module, the amount of service intervals as well as the lifetime can be positively influenced by employing appropriate filter technology. We recommend the use of commercial solutions available from industry-specific manufacturers.
- The temperature of the incoming sample gas must be at the same temperature as the temperature of the measurement cell. Deviations will create a decrease in performance.

2.6 Calibration / calibration verification / offset and span

The calibration is factory set. When the sensor module is used in conditions defined for calibration (operating temperature 20°C, pressure 1013 hPa and humidity 45% r.H.) there is no need for re-calibration. However,

an additional off-set and span can be added, for instance, to compensate for systematic errors in gas delivery and calibration equipment, or when the device is operated at a different temperature.

ⓘ If adjustment of calibration is desired, proceed as follows:

The device is designed and calibrated for measurements of a target gas in N_2 with residual humidity content.

Before executing offset and span calibration or verification, ensure the gas measurement is stable.

Allow for sufficient time for the instrument to de-gas before fixing the 0-concentration and span calibration.

First, carry out the offset and follow with the span correction procedure. To verify the calibration, the procedure can be repeated.

ⓘ The values for span and offset can be set in the Customer Frontend in the menu "Settings" -> "Sensor Parameter", changed parameter can be saved in EEPROM with "OK" button

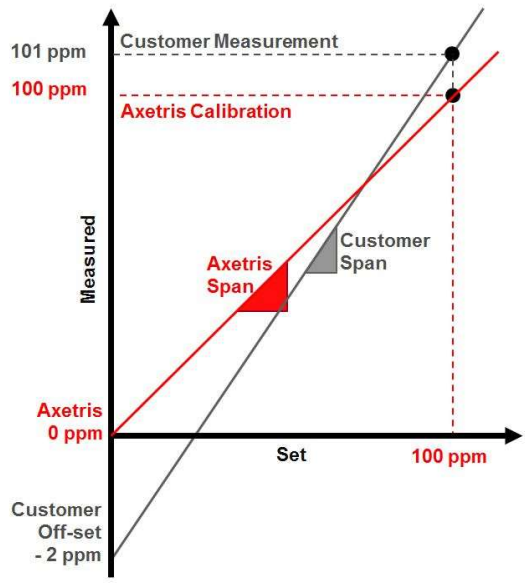


Figure 19 Schematic representation of offset and span adjust for calibration correction

Adjustment of the sensor calibration "Concentration_Offset" [ppm] and "Concentration_Span" [ppm] allows correcting the measured values with the linear function:

$$C_{new} = (C_{meas.} + Offset) \cdot Span \qquad (or \quad y = (x + b) \cdot m)$$

In practice, the correction values are entered: e.g., if the measured offset is -1 ppm, 1 has to be entered under Customer offset, and if the slope for the span (or "gain") is too steep by 5%, enter 0.95 under Customer Span to correct the slope. Upon "OK" the corrected values will be saved in the firmware of the sensor and are also used to correct the analog output.

It is best to proceed stepwise: first, correct the offset value at zero concentration and then correct for span at the application concentration (e.g. 50 ppm).

To go back to the standard calibration, the following values need to be entered: Offset = 0 / Span = 1

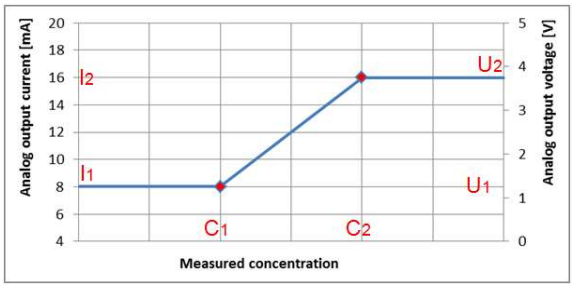
ⓘ Please note that if you have a two gases system like CH4/CO2 or CH4/C2H6 you also can correct offset and span for the 2nd gas concentration.

2.7 Analog output adjustment

The analog output of the sensor follows the measured and transmitted concentration. To define the analog output, it is important to keep in mind that the system internally operates with an analog range between 0...1. Then the electronics drive this range to either 0…5 V or 4…20 mA.

The first part of the calculation is to normalize your desired output values by your physical unit.

Let U_1 or I_1 be the voltage or current you want to get at the concentration. Let C_1 and U_2 or I_2 the voltage or current at the concentration or C_2. Then N_1 is the analog output value of U_1 or I_1, normalized by its physical unit and N_2 is the normalized value of U_2 or I_2. C_{min} is the lower value of the pair C_1 and C_2 and C_{max} is the higher one.



System with 0…5V:

$$U_{out} = (C_{meas.} \cdot Span + Offset) \cdot 5V$$

$$N_1 = \frac{U_1}{5V} \qquad N_2 = \frac{U_2}{5V}$$

System with 4…20 mA:

$$I_{out} = (C_{meas.} \cdot Span + Offset) \cdot 16mA + 4mA$$

$$N_1 = \frac{I_1 - 4mA}{16mA} \qquad N_2 = \frac{I_2 - 4mA}{16mA}$$

$$Span = \frac{N_1 - N_2}{C_1 - C_2} \qquad Offset = -C_{\min} \cdot Span + N_1$$

The span can either be positive (a rising concentration leads to a rising voltage or current) or negative (a rising concentration leads to a falling voltage or current). Now you can transmit the calculated parameters to the sensor.

- AnalogOut_Min_Conc: C_{min}
- AnalogOut_Max_Conc: C_{max}
- AnalogOut_Offset: Offset
- AnalogOut_Span: Span

You will find additional examples in the appendix of this manual in Annex B.

ⓘ If you correct offset and span in the customer calibration, it is linked to the analog output. Please adapt the customer calibration first, so that the measured concentration fits to your set points. And then adjust the analog output of your system.

ⓘ The 0 - 5V analog ouput is slightly limited due to the hardware. In reality, the 0V is around 0.1V and the 5V round 4.9V. To ensure good performance in the upper and lower range, an additional analog output calibration is recommended.

ⓘ The values for the analog output can be set in the Customer Frontend

2.8 Zeroing and Purging

Before starting a measurement, it is recommended to perform a zeroing (without target gas in the flow) and adjust the offset value accordingly. This operation is done to adapt the reading of device to the actual operation conditions (pressure, temperature, humidity …).

In case of air being used for the zeroing, make sure your device is not measuring CO2 as air may contain few hundreds ppm of CO2 that will lead to a wrong value for the offset.

2.9 Gas supply

The piping system surrounding the device is important. Leaks in the piping system can falsify the results. It is therefore recommended to check the system for leaks. The use of correct fittings and sealing helps to avoid leaks. Use standard pressure reduction equipment on gas bottles. Maximum flow is 3 l/min and minimum flow 0.1 l/min.

ⓘ When no gas tubes are connected to the sensor module, or the device is on stock the gas connector plugs should be used to prevent the gas cell from contamination.

⚠ When measuring potentially dangerous concentrations of toxic gases (CH4, C2H2, etc.), properly vent the exhaust of the device to a safe area. Improper venting of the exhaust may cause severe injury.

2.10 Adjusting firmware integration time (T90)

Depending on the application the device sends one or two values per second. The integration time can be increased for applications where longer measurement periods can be accepted. This reduces the noise of the measurement but increases the reaction time. If the integration time is increased, the measurement values sent by the device represent a moving average. The averaging used is an exponential weighted average (1-pole *IIR filter* like an RC filter, for further information check LGD FAQ document). The value set in the user interface will be saved in the firmware of the device and thus is available even after disconnecting the device.

Min value: 0 s (no Integration)

Max value: 600 s (limited to this value)

ⓘ The integration on the concentration values is linked with the analog output. The measured concentration is directly transferred into an analog value.

ⓘ The values for integration time can be set in the Customer Frontend.

3 INSTRUCTIONS FOR LGD SYSTEM INTEGRATION

ⓘ Please also check for relevant information in Chapter 2 regarding system setup for evaluation.

3.1 Integration options

To facilitate the integration of the LGD Compact-A sensor module the electronics housing as well as the interface board can be removed, and the module can be operated with the main board only. In this case, an appropriate shielding and EMC conformity has to be fulfilled by the integrating Original Equipment Manufacturer it.

Important: EMC conformity will only be granted for product executions with shielded cable versions together with electronic box.

When integrating the LGD Compact-A without electronics housing and without the interface board, the integrating Original Equipment Manufacturer must ensure an appropriate shielding, to avoid signal interferences resulting in decrease of measuring performance. The electrical connections regarding shielding described in the following sections have to be implemented.

3.1.1 With electronics housing

The shielding of the electronics and the LGD Compact-A sensor module Generation 1 is realized as shown in Figure 20

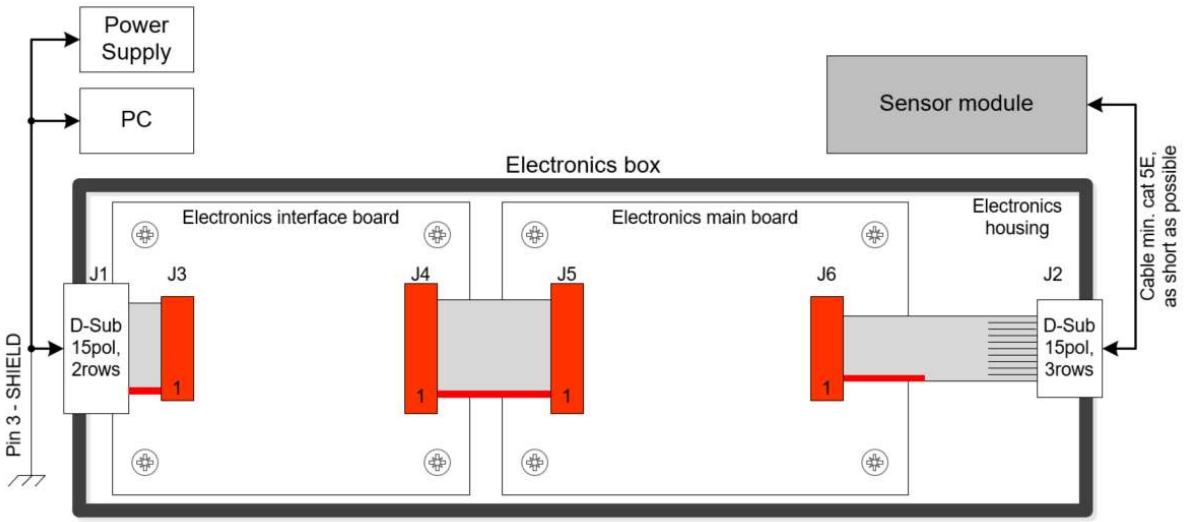


Figure 20 Shielding concept with electronics housing

Remark: Generation 2 has a stappled boards inside the housing, but connections are the same

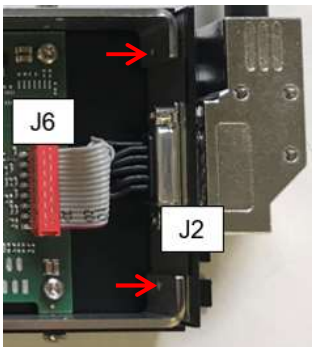


Figure 21 shielding via mounting screws of electronic housing

The shielding connection of the electronics boards and the sensor module is done over the electronics housing.

Note that the metallic cover that incorporates the D-Sub connector J2 is partly anodized. For proper shielding, it is therefore important that its mounting screws are tightened to ensure a good electrical interconnection between the sensor module and the electronics housing.

Remark: For Generation 2, J2 connector is the same (with same pin connection) where the D Sub connector is slightly different (straight)

Figure 22 shielding via mounting screws of electronic housing

The shielding connection of the electronics boards and the sensor module is done over the electronics housing.

3.1.2 Without housing

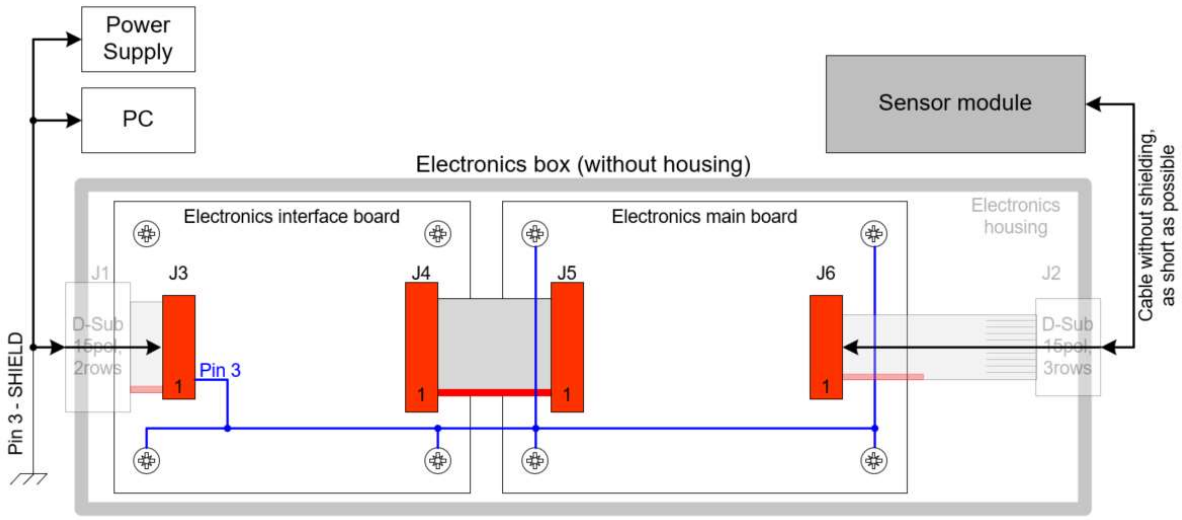


Figure 23 Ground concept without housing

Notes:

- The cable shield from the sensor module has to be connected to the earth potential.
- The six marked screws (see Figure 23) have to be connected to the same earth potential.

3.1.3 Without housing and without interface board

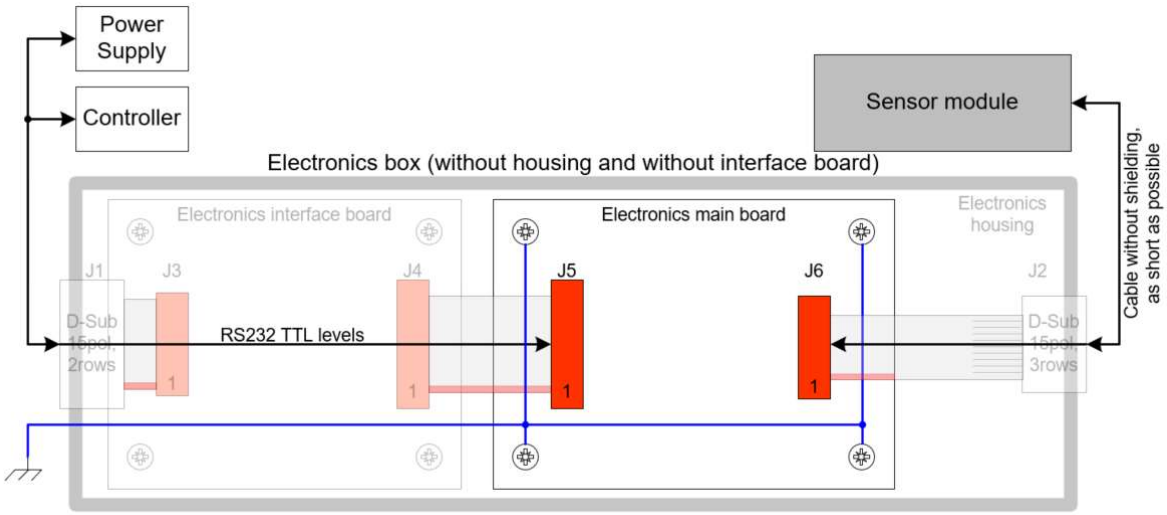


Figure 24 Ground concept without housing and without interface board

Notes:

- The cable shield from the sensor module has to be connected to the earth potential.
- The four marked screws (see Figure 24) have to be connected to the same earth potential.

3.1.4 Instructions for cable packaging

The LGD compact module Generation 1 will be delivered with an electrical cable from sensor housing to electronic boards.

The EMC conformity tests were done with the cable packaging M12 straight D sub 200 mm shielded (Phoenix Contact 1430200).

Important: EMC conformity will only be granted for product executions with shielded cable versions together with electronic box.

If this shielded cable (straight or angled version) cannot be used due to integration issues, please be aware of the following recommendations for the cable packaging:

- Conductor cross section 17x 0.14 mm² (signal line), AWG 26
- Fully shielded cable with good shield connection
- Max. cable length = 350mm (in general as short as possible)
- Twisted wire couples according to the following diagram:

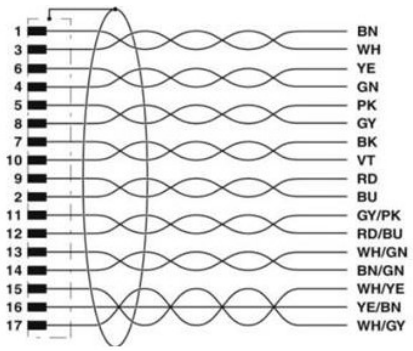
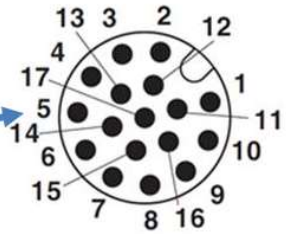


Figure 25 Diagram of the cable wiring

ⓘ Please be aware that the LGD performance is specified regarding our shielded cable versions and any other cable packaging may create a loss in performance or an increasing noise level.

Schematic diagram
Pin assignment M12 male connector, 17-pos., male side view



Connector M12	
1	Laser_Drive
2	Laser_Thermistor
3	Laser_GND
4	Laser_PT1K
5	Laser_Supply3V3
6	Laser_ID
7	Laser_TEC-
8	Laser_Udiff
9	Laser_GND
10	Laser_TEC+
11	Photodiode_I
12	Photodiode_GND
13	Gascell_PT1K
14	Gascell_GND
15	Reserve1
16	Reserve2
17	Not connected

Figure 26 Pin connection of the LGD compact sensor module Generation 1

For the Generation 2, the cable is not detachable from the sensing module and the delivered cable is shielded, fulfilling EMC conformity

Figure 27 cable/connector mounting on EMC Conform execution for Generation 2

3.2 Data transmission protocol

The LGD module has multiple commands to communicate with other devices, like microcontrollers, or with the LGD Customer Frontend when it is connected to a PC. The following describes the structure of all commands necessary for the integration and use of the sensor.

3.2.1 RS232 Serial settings

Default parameters for the serial communication to the sensor:	Baud rate	9600 (19200, 115200)
	Data bits	8
	Parity	no
	Stop bit	1
	Flow control	no

3.2.2 Overall communication structure

In the following description we will use *Command,* describing the transfer from PC to the device. Conversely, the name *Response* will denominate the equivalent transfer from the sensor to the PC.

The *Command* or *Response* IDs consist of the following:

Command / Response ID	Abbreviation
Ping	*P*
Config	*C*
Start Measurement	*M*
Idle	*I*
Save to EEPROM	*S*
Version.	*V*
Status monitoring	*D*

The data transfer is based on packets constituted of a multiple of 8 Bytes. The general *Command* structure for data transmission to the device is given in Table 7 and an example is given in Table 8.

Table 7 General *Command* structure for data transmission to the device

Byte	Value	Description
1	"{"	Start of the packet
2	{ P C M I S V}	Command ID
3	0x08	Size of the packet LSB (least significant bit)
4	0x00	Size of the packet MSB (most significant bit)
5	0xXX	Extended command and parameter
	0x00	Additional zeros to reach multiple of 8
n -1	Check sum	Check sum (sum from 1 to n-1= 0)
n	"}"	End of the packet

Table 8 Example of a *Command* to transmit data to the device (Ping user calibration)

Command in Hex	7B 50 08 00 10 00 1D 7D
Command in ASCII	{ P . . . . . . }
Description	Start of packet { Command (P = ping) Packet size (here 8 bytes) Extended command and Parameter (here 10) Additional zeros to reach multiple of 8 Checksum End of packet }

Analog to the *Command* structure, the *Response* packets have the same general format and are given in Table 9, with an example in

Table 10.

Table 9 General structure of the device *Response*

Byte	Value	Description
1	“{“	Start of the packet
2	{S P C M I V}	Command ID
3-4	N=Size of the packet	Size of the packet (in bytes)
5	Error status Flags	[E W x x x x x x] Bit_0: Error Flag; Bit_1: Warning Flag.
6	Error status Code	See Annex A and B
		Data
		Additional zeros to reach multiple of 8
n - 1	Check sum	Check sum (sum from 1 to n-1 = 0)
n	“}”	End of the packet

Table 10 Example of a *Response* from the device (Ping user calibration)

Response in Hex	7B 50 58 00 00 00 05 03 0B 00 03 0C 00 05 16 00 05 17 00 05 18 00 00 00 00 00 00 00 80 3F 00 00 FA 43 00 00 7A 44 CD CC CC 3D 00 00 00 00 00 00 00 00 00 17 B7 D1 38 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 80 7F 00 00 00 3F 00 00 80 7F 00 00 00 00 00 00 00 00 F7 7D
Response in ASCII	{ P X . . . . . ? . . . C . . z D . . . = ... 8 . . . . . ? . . . . .

be resent from the start. If the flux is consistent with the timeout requirements, the 64 bytes are read, checked for consistency and only considered if all conditions are fulfilled.

Very long commands > 64 bytes

For very long commands (> 64 bytes) it is important that the packet transmission lasts a certain amount of time. AXAG recommends adding a waiting time of 5 ms every 8 bytes during transmission, to lengthen the transmission time and avoid problems. Every valid transmission to the device will be acknowledged.

3.3 Measurement data format

After connecting the power supply, the device sends a version packet (containing 64 bytes) and then starts sending measurement packets continuously, roughly every second. All float and integer values have the little endian format (according to Norm IEEE 754).

For different device types the measurement packets have the following format:

Table 11 For Applications with 1 target gas (One measurement packet contains 16 Bytes)

Bytes (Hex)	Description	Data Type
7B 4D	Begin of packet: open brace {M	2 Bytes
10 00	Packet size = 16	Unsigned Integer 16 Bit
00 00	Error code	2 Bytes
28 54 EE 45	Concentration gas 1 = 7626.52	1 Float (Single Precision)
00 00 00 00	zeros to reach packet size multiple of 8	4 Bytes
79	Checksum	1 Byte
7D	End of packet: close brace }	1 Byte

Table 12 For Applications with 2 target gases (One measurement packet contains 24 Bytes)

Bytes (Hex)	Description	Data Type
7B 4D	Begin of packet: open brace {M	2 Bytes
18 00	Packet size = 24	Unsigned Integer 16 Bit
00 00	Error code	2 Bytes
05 71 47 42	Concentration gas 1 = 49.9 ppm	1 Float (Single Precision)
03 BC 4D 46	Concentration gas 2 = 13167 ppm abs. humid.	1 Float (Single Precision)
E0 AB F4 41	Temperature [°C] = 30.6 °C	1 Float (Single Precision)
00 00 00 00	zeros to reach packet size multiple of 8	4 Bytes
0F	Checksum	1 Byte
7D	End of packet close brace }	1 Byte

ⓘ Floats are coded based on IEEE754 - tools can be used under http://www.binaryconvert.com/convert_float.html to convert them.

(Bytes order should be reverted "Little endian.")

3.4 Change of user parameters

To change the user parameters of the device, follow the procedure as given below:

Send *Idle* (I) command

The device enters *Idle* mode and then is able to access the user data in the RAM (see Chapter 3.5.2)

Ping (P) user data

With the *Ping* command the device sends a ping packet with the user parameters from the RAM (see Chapter 3.5.3).

Configure (C) user data

The new values now can be sent to the device with a *Config* command (see Chapter 3.5.4).

Save (S) to EEPROM

To save the new parameters to EEPROM (permanently saved) please send the *Save* command (see Chapter 3.5.5).

Start measurement (M)

Start the *Measurement* again to go back to normal measurement mode (see Chapter 3.5.6).

ⓘ It is necessary to write all the user parameters in one packet. If just one value needs to be changed, ping all values from the device. Then overwrite the desired value and write all the values back to the device.

ⓘ If user data are not saved to EEPROM they will be lost after a reboot of the system.

ⓘ The write cycles of an EEPROM are limited depending on external conditions. Under unfavourable conditions the write cycles might be limited to 10'000 only.

3.5 Commands

3.5.1 Version

This command is used to obtain the version of the firmware from the device. It has a length of 8 bytes and there are no further parameters that can be added to this command.

Command in Hex	7B 56 08 00 00 00 27 7D
Command in ASCII	{ V . . . . ' }

The response (64 bytes) contains the device serial number and diverse checksums.

Table 13 Explanation of the Version-response

Byte	Value	Description
7-22	String (16 bytes)	Firmware version
23-54	String (32 bytes)	Serial number of the system
55-56	Integer 16 (2 bytes)	Checksum firmware
57-58	Integer 16 (2 bytes)	Checksum tools
59-60	Integer 16 (2 bytes)	Checksum config
61	Integer 8 (1 byte)	Checksum hardware
62	Integer 8 (1 byte)	Checksum update

ⓘ After power-on the device will send this *Version*-packet only once. This also indicates a start up / reset.

3.5.2 Stop measurement (Idle)

Send the *Idle*-command to stop the current measurement. This allows accessing (*Ping*) the user parameters.

Command in Hex	7B 49 08 00 00 EA 4A 7D
Command in ASCII	{ I . . . . J }

As a confirmation, the device responds with an *Idle* (without any data). The device now is in *Idle*-mode (standby) and waits for further commands. It will not send any further data or packages.

3.5.3 Ping user data

With the *Ping* command user data can be asked from the device. Please note that the device needs to be in *Idle*-mode to have access to the user data.

Command in Hex	7B 50 08 00 10 00 1D 7D
Command in ASCII	{ P . . 0 . . }

The device transmits a *Ping*-packet with the following content (content can vary depending on product execution):

Table 14 *Response* from the device with description of the bytes

Byte	Possible HEX	Description	
1 – 4	7B 50 50 00	Start of packet with size (80 bytes)	
5 – 6	00 00	status code bytes	
7 – 10	00 00 40 40	Concentration gas 1 Offset	(here 3)
11 – 14	CD CC 8C 3F	Concentration gas 1 Span	(here 1.1)
15 – 18	00 00 00 00	Reserved variable for future use (dummy)	
19 – 22	00 00 00 00	Reserved variable for future use (dummy)	
23 – 26	00 00 00 00	Analog output: Min concentration	(here: 0)
27 – 30	00 00 C8 42	Analog output: Max concentration	(here: 100)
31 – 34	00 00 00 00	Analog output: Offset	(here: 0)
35 – 38	0A D7 23 3C	Analog output: Span	(here 0.01)
39 – 42	00 00 20 41	Integration time	(here: 10)
43 – 46	00 80 BB 44	Concentration gas 2 Offset	(here 1500)
47 – 50	33 33 73 3F	Concentration gas 2 Span	(here 0.95)
51 – 54	00 00 00 00	Diagnostic Mode (1 = active / 0 = inactive) For further details about Diagnostic mode, contact Axetris AG.	(here 0)
55 – 58	00 00 00 00	Reserved variable for future use (dummy)	
59 – 62	00 00 00 00	Reserved variable for future use (dummy)	
63 – 66	00 00 00 00	Reserved variable for future use (dummy)	
67 – 70	00 00 00 00	Reserved variable for future use (dummy)	

71 – 74	00 00 00 00	Reserved variable for future use (dummy)
75 – 78	00 00 00 00	Trailing zeros
79 – 80	7F 7D	End of packet with checksum

ⓘ Please note that this general description may be subject to changes depending on specific product execution.

The detailed list of the available user data for a specific product execution can be found in the LGD Customer Frontend under "Settings" -> "Sensor Parameter"

3.5.4 Configure user data

The *Config* command is used to change the user data of the system. The parameter of this command is the list of the user data.

Table 15 Configure user data command with description of the bytes

Byte	Possible HEX	Description	
1 – 4	7B 43 50 00	Start of packet with size (80 bytes)	
5 – 6	00 EA	Additional command	
7 – 10	00 00 40 40	Concentration gas 1 Offset	(here 3)
11 – 14	CD CC 8C 3F	Concentration gas 1 Span	(here 1.1)
15 – 18	00 00 00 00	Reserved variable for future use (dummy)	
19 – 22	00 00 00 00	Reserved variable for future use (dummy)	
23 – 26	00 00 00 00	Analog output: Min concentration	(here: 0)
27 – 30	00 00 C8 42	Analog output: Max concentration	(here: 100)
31 – 34	00 00 80 BF	Analog output: Offset	(here: -1)
35 – 38	0A D7 23 3C	Analog output: Span	(here 0.01)
39 – 42	00 00 20 41	Integration time	(here: 10)
43 – 46	00 80 BB 44	Concentration gas 2 Offset	(here 1500)
47 – 50	33 33 73 3F	Concentration gas 2 Span	(here 0.95)
51 – 54	00 00 80 3F	Diagnostic Mode (1 = active / 0 = inactive) For further details about Diagnostic mode, contact Axetris AG.	(here 1)
55 – 58	00 00 00 00	Reserved variable for future use (dummy)	
59 – 62	00 00 00 40	Reserved variable for future use (dummy)	
63 – 66	00 00 00 00	Reserved variable for future use (dummy)	
67 – 70	00 00 00 00	Reserved variable for future use (dummy)	
71 – 74	00 00 00 00	Reserved variable for future use (dummy)	
75 – 78	00 00 00 00	Trailing zeros	
79 – 80	A2 7D	End of packet with checksum	

As confirmation, the device responds with a *Config*-packet without any data.

ⓘ Please note that this general description may be subject to changes depending on specific product execution.

3.5.5 Save to EEPROM

Changing the user parameter just stores them in the RAM. To save them from RAM to EEPROM use the save command.

Command in Hex	7B 53 08 00 40 EA 00 7D
Command in ASCII	{ S . . @ . ! }

As a confirmation, the device responds with a *Save*-packet without any data.

ⓘ The write cycles of an EEPROM are limited depending on external conditions. Under unfavourable conditions the write cycles might be limited to 10,000 only.

3.5.5.1 Changes in Measurement Mode without Save to EEPROM

The switch from single gas to dual gas measurement for the execution LGD Compact CH_4/C_2H_6 is normally done several times a day during a normal operation in the field. Using the Save to EEPROM command after every change would shorten the lifetime of the sensor module seriously due to the limited writing cycles of the EEPROM.

The change from single gas to dual gas measurement can be done simply by sending the "Idle" command following by the "Configure" command (3.5.4) and starting the measurement. The changes will be kept in the configuration until the next power disruption of the sensor module.

ⓘ For the execution LGD Compact CH4/C2H6 it is important not to send Save to EEPROM command when changing the Measurement Mode from single gas to dual gas measurement to avoid a shorter a lifetime due to limited writing cycles of the EEPROM.

3.5.6 Starting measurements

If the device is in *Idle*-mode and does not perform measurements, it can be re-started with the *Measurement*-command.

Command in Hex	7B 4D 08 00 01 EA 45 7D
Command in ASCII	{ M . . . . E }

As a confirmation, the device responds with a *Measurement*-packet without any data.

ⓘ The device then starts to initialize the measurement procedure and continuously sends measurement packets with the measured gas concentrations and other measured values, depending on the product type. These packets also contain M as a measurement response.

3.6 Status, error and failure response

Additionally, the device also can send special packets regarding errors (E) or failures (F). During the measurement cycles it can transmit separate error or failure packets. Both do not contain any further data.

Error-Packet:

Table 16 Error packet sent from device (Low power on the photodiode)

Response in Hex	7B 45 08 00 CE F4 76 7D
Response in ASCII	{ E . . . . 5 }

The *Error*-packet contains status, warning, and error codes.

Failure-Packet:

Table 17 Failure packet sent from device (Error checksum not correct)

Response in Hex	7B 46 08 00 C1 0F 67 7D
Response in ASCII	{ F . . . . g }

The *Failure*-packet contains error codes that require manufacturer-assisted maintenance of the device.

ⓘ While working with the Axetris Customer Frontend, the device sends additional status information packets ({D…}). The D-packets are for internal use only and can only be monitored while working with the Axetris Customer Frontend (they can be ignored by the customer in standard device operation). When quitting the Axetris Customer Frontend to work with the LGD module Communication Protocol, the device stops sending D-packets.

4 MAINTENANCE AND TROUBLE SHOOTING

4.1 Maintenance

The Axetris LGD Compact gas sensor OEM-modules will not require routine maintenance when the installation and operation instructions described in this document are carefully applied. The surface of the instrument can be cleaned with a soft tissue and the use of isopropyl alcohol. Any gas filtration and conditioning equipment must be checked and replaced periodically according to manufacturer specifications.

If the measurement cell gets contaminated or damaged, please contact Axetris for cell maintenance and replacements. As technical training is necessary for this procedure, we recommend setting up a system exchange program. Please contact Axetris for details.

In case of problems, re-check all the recommendations presented in this document and be sure they have been followed and fully applied. If the observed problems persist, contact your Axetris sales person.

ⓘ It is important that adequately trained and qualified personnel only service this device.

ⓘ This instrument contains electronic components that might be damaged by static electricity. The device cover should never be removed as it might damage the optics and/or electronics.

⚠ Axetris gas sensor modules contain metal and elastomeric sealing parts. The customer is responsible for checking the compatibility of those materials with the gas applied to the device. Not observing these recommendations might lead to personal injury or death.

⚠ It is recommended that the user checks the device regularly to ensure that it is leak-free. Sealings may change their characteristics with age, exposure to process gas, temperature and pressure.

⚠ If it is necessary to remove the device from the system, unplug the electrical power before any other manipulation.

⚠ Before removing the device from the system, purge thoroughly with a dry inert gas such as Nitrogen before disconnecting the gas fittings. Ensure that the piping is de-pressurized. Failure to correctly purge and de-pressurize the device could result in fire, explosion, or severe injury.

4.2 Troubleshooting

If you have problems while using the LGD Customer Frontend, please check if

- The device' power supply is plugged-in and properly working
- The serial connection is set and data transmission is working
- The correct serial port is set on start-up

ⓘ If the LGD Compact measurements do not correspond to the expected values, please check that the sensor is used within the specified operation and calibration range.

After verifying the factors above, follow the troubleshooting procedures outlined here below:

Problem	Possible cause	Solution
No signal / No value output	No power supply or wrong polarity	Check wiring
No reaction when target gas is flowed in	Condensation in system	Flow with dry and dust-free air in the gas inlet for at least 15 min
Every 10 or 11 acquisition points the time between points is longer	When the target gas concentration is below a certain limit, the device has an internal check which lasts longer (around 2s)	Flow with higher amount of target gas
The T90 is longer than expected	The flow is lower as 0.1 l/min	Increase the flow rate

5 IMPORTANT DISCLAIMER / NOTICE

The information furnished by Axetris AG is believed to be correct and accurate. However, Axetris shall not be held liable to recipient of any third-party for any damages, including but not limited to personal injury, property damage, loss of profits, loss of use, interruption of business or indirect, special incidental or consequential damages, of any kind, in connection with or arising out of the furnishing, performance or use of technical data provided herein. No obligation or liability to recipient or any third-party shall arise or flow out of Axetris rendering of technical or other services.

6 FCC STATEMENT

This equipment has been tested and found to comply with the limits for a class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interferences in a residential installation. This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

ANNEX

A. Status, Warning and Error Code Listing for Evaluation

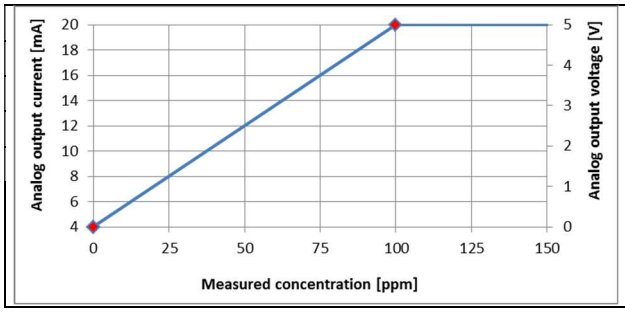
Error code (dec.)	Error code (hex.)	Description	Action / Solution
223	0xDF	PeakCheck process: Sensor was not able to perform PeakCheck correctly	Start PeakCheck process again Check conditions as described in QuickGuide
224	0xE0	PeakCheck process: C2H6 might not be detected properly.	Try with lower CH4 concentration
225	0xE1	PeakCheck process: Gas2 peak position out of range	Service required
226	0xE2	PeakCheck process: Gas2 peak position near limit	Service required soon (Warning only)
227	0xE3	PeakCheck process: Sensor is performing PeakCheck. Do not interrupt	NA
228	0xE4	PeakCheck process: Distance between gas1 and gas2 out of range	
238	0xEE	Sensor line-locking not secured	Contact Axetris customer support if problem persists
239	0xEF	End of life	The sensor will soon be out of service
240	0xF0	Opticalhead's temperature out of range	Check if ambient temperature is within specified range. If yes, there is a system failure. Service is required.
241	0xF1	Laser driver failure. Sensor not measuring	Check if ambient temperature is within specified range. If yes, there is a system failure. Service is required.
242	0xF2	Laser driver failure. Sensor not measuring	Service required
243	0xF3	Low power on the photodiode. Less than 10% of initial power	Check for condensation. Otherwise service is required
244	0xF4	Low power on the photodiode. Less than 50% of initial power. Sensor is functioning but with compromised performances due to misalignment, dirt or condensation.	Service required soon (Warning only)
245	0xF5	Too high power on photodiode. More than 125% of initial power. Sensor functionality compromised	Service required
246	0xF6	Laser temperature compensation executed (system restarting)	Laser near its EOL
247	0xF7	Laser current compensation limit reached	EOL reached. Service required
248	0xF8	Cell temperature is out of safe range. Sensor not measuring	Check ambient temperature Ensure proper venting
249	0xF9	System temperature out of range (-40°C … +80°C) Sensor not measuring	Check ambient temperature Ensure proper venting
250	0xFA	Laser driver failure Sensor not measuring	Check ambient temperature. If ok, service required

251	0xFB	Laser driver failure Sensor not measuring	Service required
252	0xFC	Warm-up phase OR out of calibration temperature range System performance could be compromised	Check ambient temperature
253	0xFD	Laser temperature compensation executed (system restarting)	Laser near its EOL
254	0xFE	Alarm 1 out of defined limit Customer & application specific parameter	Sensor working correctly. User can change Alarm 1 level
255	0xFF	Alarm 2 out of defined limit Customer & application specific parameter	Sensor working correctly. User can change Alarm 1 level

B. Examples for Analog Output Adjustments

See Chapter 2.7 for instructions.

Example 1 [0 ppm = 4 mA or 0 V; 100 ppm = 20 mA or 5 V]



	min	max	delta
Concentration [ppm]	0	100	100
Analog voltage [V]	0	5	5
Analog current [mA]	4	20	16
Normalized (N_1; N_2) []	0	1	1

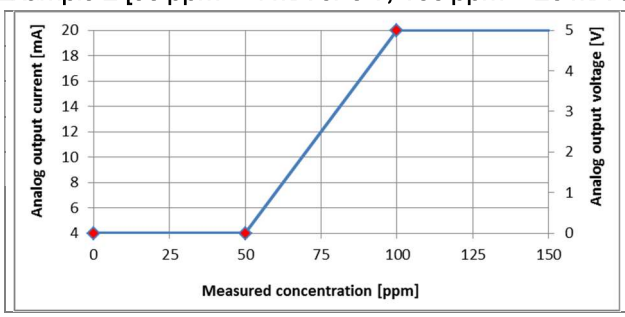
$$Span = {}^{1}\!/_{100ppm} = 0.01$$

$$Offset = 0 \cdot 0.01 + 0 = 0$$

For this example, the sensor parameters to type into the interface are:

- AnalogOut_Min_Conc: 0
- AnalogOut_Max_Conc: 100
- AnalogOut_Offset: 0
- AnalogOut_Span: 0.01

Example 2 [50 ppm = 4 mA or 0 V; 100 ppm = 20 mA or 5 V]



	min	max	delta
Concentration [ppm]	50	100	50
Analog voltage [V]	0	5	5
Analog current [mA]	4	20	16
normalized []	0	1	1

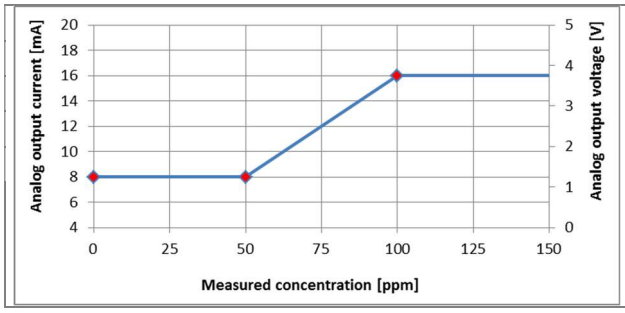
$$Span = {}^{1}\!/_{50ppm} = 0.02$$

$$Offset = -50 \cdot 0.02 + 0 = -1$$

For this example, the sensor parameters to type into the interface are:

- AnalogOut_Min_Conc: 50
- AnalogOut_Max_Conc: 100
- AnalogOut_Offset: -1
- AnalogOut_Span: 0.02

Example 3 [50 ppm = 8 mA or 1.25 V; 100 ppm = 16 mA or 3.75 V]



	min	max	delta
Concentration [ppm]	50	100	50
Analog voltage [V]	1.25	3.75	2.5
Analog current [mA]	8	16	8
normalized []	0.25	0.75	0.5

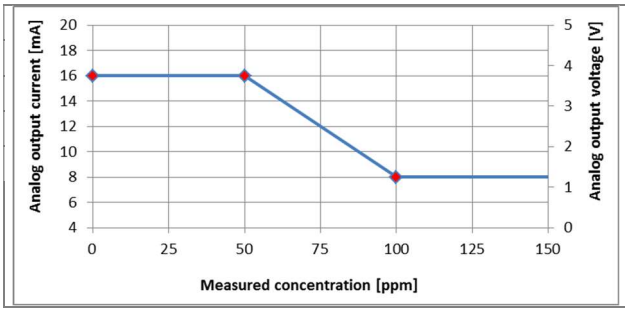
$$Span = {}^{0.5}\!/_{50ppm} = 0.01$$

$$Offset = -50 \cdot 0.01 + 0.25 = -0.25$$

For this example, the sensor parameters to type into the interface are:

- AnalogOut_Min_Conc: 50
- AnalogOut_Max_Conc: 100
- AnalogOut_Offset: -0.25
- AnalogOut_Span: 0.01

Example 4 [50 ppm = 16 mA or 3.75 V; 100 ppm = 8 mA or 1.25 V]



	min	max	delta
Concentration [ppm]	50	100	50
Analog voltage [V]	3.75	1.25	-2.5
Analog current [mA]	16	8	-8
normalized []	0.75	0.25	-0.5

$$Span = -0.5 / 50ppm = -0.01$$

$$Offset = -50 \cdot (-0.01) + 0.75 = 1.25$$

For this example, the sensor parameters to type into the interface are:

- AnalogOut_Min_Conc: 50
- AnalogOut_Max_Conc: 100
- AnalogOut_Offset: 1.25
- AnalogOut_Span: -0.01

Name of Authorized document official: Rui Protasio, Document Owner

Disposal

Electrical equipment, accessories and packaging should be recycled in an environment friendly way. When you are disposing of our products, please observe the national and local regulations.

For EU countries: Do not dispose of electrical equipment with household refuse.